With receipts: twelve checked cases from agent-assisted audits of palaeobiology's data and software

Barry W. Brook*[1], Claude (Anthropic) and OpenAI Codex[2]

Internal research draft from a living repository. Release and submission are Barry W. Brook's decision.

1. University of Tasmania, Hobart, Tasmania, Australia
2. AI research collaborators; roles are stated in "Roles of AI systems and humans"

Corresponding author: Barry W. Brook

Abstract

Quantitative palaeobiology rests on shared infrastructure: occurrence databases, published data deposits, time scales, plate models and analysis packages. Problems in these layers can pass into published inference unnoticed. During the first month of a research programme run by two AI systems (Anthropic's Claude and OpenAI's Codex) with a human facilitator, agent-assisted audits produced candidate findings about this infrastructure. We report twelve as a case series. Each was chosen for its consequence for inference, replayed from pinned inputs by a different actor, and checked by the computational or mathematical routes recorded in its receipt. The cases include:

- deposited occurrences absent from a dated public database snapshot (71,103 of 368,760 molluscan records);
- interval ages that change between archived database snapshots;
- excess rows in a database update that changed 1,388 of 2,267 exposure means in one of our own analyses;
- stored sample ages that disagree with their own tie points;
- three implementation defects in analysis packages;
- a preservation-rate calibration failure, reported beside retained positive controls;
- an extinction-time interval that measures dating sensitivity only;
- model-dependent palaeolatitudes;
- two published spreadsheets that differ from their sources;
- a printed significance value that its stated test does not reproduce (0.316, not 3.4×10^{-16});
- archived calibration scores that do not replay.

Every proposition is bounded to the version, design and convention tested. Replay narrowed or withdrew several of the original statements, and we report those corrections. The series does not estimate how common such problems are, does not show that agent-assisted auditing is better or cheaper than other auditing, and does not test whether any published conclusion changes. It records, with commands and input hashes, the kinds of problem that systematic replay exposed in one programme and what each one does and does not change.

1. Introduction

A palaeobiological result is usually several layers of other people's work deep. An occurrence count depends on a database query and that database's state on the day of the query. An age depends on a time scale and an age model. A palaeolatitude depends on a plate model and its reference frame. A rate depends on an estimator and on the package that implements it. Each layer is maintained by others and changes over time.

The community has asked for scrutiny of exactly this layer. In a recent community survey of open questions in palaeontology (Smith et al. 2025), several priorities concern it: quantifying collecting and preservation biases, finding and correcting data-entry biases, accounting for data loss, and making data open and combinable. Audits of the infrastructure bear directly on those priorities.

During October and November 2026, the Extinction and Palaeontology Research Programme (EPRP) used two AI systems, with a human facilitator, to vet research questions against the data that could answer them. Vetting meant mirroring databases and deposits, benchmarking packages against known truth, and replaying published numbers. That work produced an initial catalogue of 189 candidate findings about the infrastructure itself, and later findings as the programme continued.

This paper reports twelve of them as a case series. For each case we state the checked proposition, the evidence and how it was replayed, what it changes, and what it does not show. We also report what the auditing agents themselves got wrong. The aim is not a rate or a ranking. It is a set of inspectable cases, each with a recorded replay command.

2. How the cases were found, chosen and checked

Finding. The candidates came from read-only mirrors of databases, deposits and plate models (each hashed on arrival), benchmark suites declared before they were run, and checks on published numbers made while vetting research questions. Three working sessions, for databases, packages and references, recorded each

finding with its evidence location and the checks it had received. Candidates include positive controls and intended behaviours as well as suspected problems.

Choosing. From the initial catalogue and subsequent programme findings we chose twelve cases by consequence for inference: what each would change in an analysis someone plausibly runs. Being a defect was not enough on its own. We also wanted the selection to cover the kinds of finding in the pool, and we retained positive controls. Licence and bibliographic-metadata problems were left out as having little bearing on inference. Location data whose description could itself cause harm were left out entirely. The selection note, with reserves, is in the repository (Supplement S1).

Checking. Each case was replayed from pinned inputs on 8 November 2026 by a different actor from the one who made the original finding, combined with the checks specified in its receipt. These included separate aggregation, interpolation, geometry and scoring code, recounts from raw files and closed-form calculations. Independence is check-specific. For example, the INDITEK forward run reused the original pinned harness, and only its scoring was separately implemented. Some reviewers used the same model family and shared inputs. Each proposition is a row in one claims ledger, with its value, evidence files, the commit at which the evidence was verified, a replay command, and a statement of what it does not show. Where the replay did not reproduce an original number, the number was withdrawn, and the withdrawal is recorded (Section 4; Supplement S3).

Classes. Each case is classified by what kind of finding it is:

- an implementation defect;
- a data inconsistency;
- a calibration failure under a stated model;
- a semantic finding (what a quantity actually measures);
- an intended version or model choice that users need to know about;
- a snapshot-membership or provenance finding;
- a replay failure.

These classes matter. An intended version change is not an error, and calling it one would be unfair to the people who maintain the resource.

3. The cases

Table 1. The twelve cases. Values are from the claims ledger; each row's conditions are in the text.

#	Case	Class	Checked proposition	Claims
1	Deposited records absent from a dated public snapshot	membership and provenance	71,103 of 368,760 molluscan occurrences in a published deposit are not in a public snapshot of 23 October 2026	C191, C192
2	Interval ages change between archived snapshots	intended version choice	460, 317 and 17 intervals changed modal base age across three snapshot brackets	C076
3	Excess rows in a database update	data inconsistency	28,832 excess rows under a specified key; removing 16,260 changed 1,388 of 2,267 exposure means	C043, C045
4	Stored ages disagree with their tie points	data inconsistency	4,111 of 56,387 comparable samples differ by more than 0.01 Myr	C040
5	Age–depth conversion above hiatuses	implementation defect	sparse model: 3,972 of 8,282 query points mis-converted, maximum error about 3.0 Myr	C096
6	Numbers compared as text	implementation defect	at a threshold of 5, differences of 10, 20, 49.9, 100 and 499 were not flagged	C102

#	Case	Class	Checked proposition	Claims
7	Preservation-rate calibration, with retained controls	calibration failure	intervals covered the truth in 81 of 100 and 33 of 50 replicates, shifted high	C107, C194, C196, C105, C111
8	What an extinction-time interval measures	semantic finding and defect	the interval reflects resampled dating error only; the call overwrites the caller's random-number state	C110, C098
9	Plate-model palaeolatitudes	intended model choice	median spread among three models, age dependent, from 2.9 degrees at 10 Ma to 13.6 at 400 Ma	C125, C123
10	Published spreadsheets differ from their sources	data inconsistency	2,967 of 6,562 capture histories altered; 220 rows shifted one column	C035, C048
11	A printed probability	arithmetic error in a published claim	the stated binomial test gives 0.316, not 3.4×10^{-16}	C190
12	Archived calibration scores	replay failure	three saved scores differ from rescored values by 42.4, 17.5 and 11.9	C195, C193

3.1 Data membership and provenance

Case 1. A published global analysis of marine latitudinal diversity (Benson et al. 2025) deposited its occurrence data. Of the 368,760 occurrences in the deposit's molluscan member, 71,103 are not present in a public Paleobiology Database snapshot taken on 23 October 2026. Absence from a dated snapshot does not by itself establish why the records are absent: deletion, merging and other database changes can all contribute. Rebuilding from that dated snapshot would omit these deposited identifiers. The consequence showed up in our own work. A simple presence screen for tropical support outside Europe and North America passed in 14 of 21 stages with the full deposit, and in 12 with snapshot members only. We report these records as counts only.

Case 2. Archived query products from the Paleobiology Database show interval base ages changing between consecutive snapshots. Across three main brackets, 460, 317 and 17 intervals changed their modal base age, counting intervals with at least 50 single-interval occurrences in the later snapshot. If the threshold must hold in both snapshots, the middle count is 316. The statistic is the modal base age of an interval, not every record's age. A query repeated on different dates can therefore return different ages for the same interval. We have not shown whether the database keeps historical versions accessible, and we make no comparison with the international chronostratigraphic chart. This is a version change, and the point for users is to record the date and version of every query.

Case 3. The 2023 update of the Triton planktonic-foraminifera database (Fenton et al. 2021) contains 28,832 excess rows under the key we specified. Rows that share a key are not necessarily repeated physical observations, so we call them excess rows under that key, not duplicates. A conservative recipe removes lower-priority-source rows only from cross-source groups with recorded depth, retaining every row from the highest-priority source; it removes 16,260 rows. The consequence is measured in one of our own analyses. Rebuilding EPRP's exposure index with and without the removal changed 1,388 of 2,267 comparable unrounded lineage-bin means, 411 of them by at least one degree. Of 491 lineage-pair rows, 8, 8, 2 and 1 changed their classification under four different classification rules. An earlier internal report of 1,454 and 423 changed means did not reproduce under the pinned recipe and is withdrawn (Supplement S3).

Case 4. The final snapshot of the Neptune Sandbox Berlin database (Renaudie, Lazarus, and Diver 2020) holds 61,862 samples. We recomputed ages by linear interpolation from each hole's current tie points, where those tie points permitted. 56,387 samples have both a stored and a recomputed numeric age. Of these, 4,111 in 56 holes differ by more than 0.01 Myr. The recomputation takes the first age where two tie points share a depth; with the other choice, 8 of the differences disappear. That sensitivity concerns exact ties only, not

other interpolation models. Neither set of ages is thereby shown to be right. A user who takes the stored ages and one who recomputes from the tie points will be working from different numbers.

3.2 Software correctness and semantics

Case 5. admtools 0.7.0 converts stratigraphic height to time. We tested it on one synthetic age–depth model with two hiatuses, in two versions, against exact linear-segment truth. In the sparse tie-point version, 3,972 of 8,282 query points were mis-converted, with maximum error about 3.0 Myr. In the dense version, 6 of the same queries were mis-converted, with maximum error 0.466 Myr, and a targeted probe near a hiatus gave about 2.7 Myr. The errors fall in affected interpolation segments; later segments can be correct.

Case 6. fossilbrush 1.0.6 flags occurrences whose ages differ from a reference range by more than a threshold. It compares the differences with the threshold as text. At a threshold of 5, it failed to flag differences of 10, 20, 49.9, 100 and 499, while flagging others on the grid, including 50 and 99. Character order predicts every flag on the test grid. This is a tested mechanism on a deterministic grid, not a measured rate of misflagged records in real data.

Case 8. The GRIWM estimator of extinction time (Bradshaw et al. 2012), as implemented in Rextinct 1.1.0, reports an interval whose width comes only from resampling the input dating errors. With zero dating error, the interval collapses to a point. The interval therefore measures the sensitivity of the estimate to uncertain dates, conditional on the records. It does not quantify uncertainty from the fossil-sampling process. Separately, the implementation overwrites the calling program's random-number state, so any analysis that draws random numbers after calling it inherits a stream it did not choose. We have not quantified the effect of either finding on any published study.

3.3 Calibration, with retained controls

Case 7. We simulated fossil records under a declared constant-rate, homogeneous-Poisson design and asked whether PyRate's (Silvestro, Salamin, and Schnitzler 2014) 95% intervals for the preservation rate covered the true value. They covered it in 81 of 100 replicates in one arm and 33 of 50 in a lower-preservation arm. Posterior medians sat above the truth, with median relative bias 0.0895 and 0.458. Every miss lies below the lower endpoint: the intervals are shifted high. The retained width comparisons do not support an earlier diagnosis that the intervals were too narrow. They do not establish calibrated widths or the cause of the shift.

The benchmark also retained positive controls, in a different design with a different target: range-end bounds for retained species in two simulated clades. Under homogeneous sampling, the classical Strauss–Sadler bound (Strauss and Sadler 1989) covered the truth in 1,895 and 1,899 of 2,000 cases in two implementations, and the Robson–Whitlock bound in 1,909. The G1 and G2 generator moment controls matched their analytic expectations. Two qualifications travel with these controls:

- **A declared gate failed.** The pre-declared overall agreement gate required 99% of paired bounds to agree within 0.02 Myr; 0.91 did.
- **The hiatus arm deviated from its declaration.** Its executed windows were shifted one Myr from the declared design. In that arm, Strauss–Sadler coverage fell to 1,125 of 1,272 retained cases, with 728 cases dropped.

These controls show that the benchmark did not report failure everywhere. They do not rank the classical bounds against PyRate, which target a different quantity.

3.4 Spatial models

Case 9. We reconstructed a starting pool of 300 public points at six ages in three plate models: MERDITH2021 (Merdith et al. 2021), PALEOMAP and MATTHEWS2016_pmag_ref (Matthews et al. 2016). At each age, the median spread of palaeolatitude among the three models is taken over the points reconstructable in all three; 194 points are reconstructable at every age. The median spread is age dependent and not monotonic: 2.9 degrees at 10 Ma, 5.7 at 50 Ma, 4.3 at 100 Ma, and the largest, 13.6, at 400 Ma. This is structural disagreement among models, not calibrated error, and model choice can change

the latitude-band assignments of a diversity-gradient analysis. Separately, two routes to reconstructing TorsvikCocks2017 differ by an approximately rigid, age-specific rotation. That is consistent with their using different absolute reference frames. We do not identify either frame as correct. A model name alone does not fix the coordinates.

3.5 Published deposits and published numbers

Case 10. The electronic supplement of a study of extinction selectivity (Monarrez, Heim, and Payne 2023) provides capture histories in a spreadsheet as well as an intact binary file. Comparing the two, we found 2,967 of 6,562 histories in the spreadsheet with altered values. A further 3,595 can be restored exactly by padding to the known width of 86 characters. We did not establish how the spreadsheet was produced. Separately, the one-file All-Life table of the Fossil Record 2 compilation has 220 rows shifted one column left relative to its chapter workbooks. Of these, 112 shift ranges only, 2 shift habitat and range, and 106 move a not-found-in-record flag. We did not rerun either study's analysis, and we make no claim about its conclusions.

Case 11. A recent analysis of cyclicity in stratigraphic boundaries (Rampino and Caldeira 2025) states a binomial test with n = 47, k = 28 and p = 0.55, and prints a tail probability of 3.4×10^{-16}. The stated test gives 0.316. We confirmed this by exact rational arithmetic, by integer polynomial convolution, by a direct exact sum and in R. This does not bear on the paper's separate spectral analysis, which we did not rerun.

Case 12. INDITEK-2.0 (Herrero-Gascón et al. 2026) deposits a proof example that uses synthetic observations, with archived calibration proposals. We reran three archived rejected proposals with the original pinned harness. Their prediction vectors reproduce within about 5×10^{-14} across 2,978 cells. Their saved residual sums of squares, however, differ from rescoring under the deposited observations and scorer by 42.4, 17.5 and 11.9, and a separately implemented, model-free scorer agrees with the rescored values. These are rejected proposals, not accepted draws. The cause, and any connection to published figures, are unresolved.

4. What the auditing agents got wrong

The agents' own errors are part of the record. Replay withdrew or narrowed several original statements:

- **Case 3.** The changed-mean counts of 1,454 and 423 did not reproduce under the pinned recipe, and an 11–13% sample inflation figure was withdrawn.
- **Case 4.** A sample count that included a header row was corrected to 61,862.
- **Case 5.** "Wrong above every hiatus" was false; the error is confined to affected segments.
- **Case 7.** The "too narrow" diagnosis of PyRate's intervals was not supported.
- **Case 10.** A count of 5,291 cells in scientific notation had been read as a count of damaged histories.

The draft of this paper was itself corrected in review. Among other things, the first draft joined the admtools count from one control with the maximum from another, and attributed the INDITEK archive to the wrong paper.

Earlier in the programme, the two agents corrected each other's scientific reasoning repeatedly. One agent's proposed rule, for example, was refuted by the other's counterexample, and a predicted outcome of a screening test was wrong. These errors were identified during rechecking by a different actor. The series contains no comparison that could show whether this arrangement catches more errors than any other.

5. Recommendations

- **For database providers:** expose versions and change logs for interval ages and records, and say in query results when records have been removed or merged.
- **For package authors:** test against known truth, including hiatuses and boundary cases; compare numbers as numbers; do not reset the caller's random-number state; document what each reported interval measures.
- **For authors of deposits:** deposit identifiers and histories as text in machine-readable formats, not in spreadsheets; archive the scorer and observations with any saved calibration scores.

- **For users:** record the date and version of every query; rerun spatial analyses under more than one plate model and state the reference frame; replay a published number before building on it.
- **For programmes that use AI agents:** declare tests before running them, keep failed gates, have a different actor recheck each finding, and publish reviewed correspondence.

6. Discussion

6.1 What follows

Each case is small on its own. Together they show problems of several distinct kinds in the layer beneath published palaeobiology: version changes users need to know about, inconsistencies within a resource, implementation defects, intervals whose meaning is narrower than their label suggests, and published numbers that do not survive replay. None needed new data or new theory to find. Each needed a layer to be replayed against known truth or against its own source.

6.2 Which conclusions depend on which assumptions

Table 2. Dependence of the main conclusions on their assumptions.

Conclusion	Rests on	Fails or narrows if
Each case's proposition holds for the version tested	Pinned inputs and the stated convention	The replay is wrong for those inputs; another version only restricts scope
The measured changes are real	The support counts, inputs and EPRP analyses named in each case	The named downstream calculation was misreplayed; effects on published conclusions were not tested
The checks add information	A different actor and the routes recorded in each receipt	The routes share an error, such as a common input misread
Provider knowledge	Not established	Open: providers may already have documented a behaviour

6.3 What would sharpen the answer

- **Disclosure responses.** Providers' and authors' replies would establish which findings are already known, intended or fixed.
- **A controlled comparison.** The same audit run by experienced human auditors would allow the cost and yield of agent-assisted auditing to be compared, which this series cannot do.
- **Downstream replays.** Rerunning published analyses that use the affected layers would measure consequence for published conclusions directly.
- **A cold public rebuild.** Rebuilding every case from public inputs alone would test how far the recorded commands suffice without locally retained images and archives.

7. Conclusions

Systematic replay of palaeobiology's shared infrastructure, in one programme, exposed problems of several kinds. The twelve cases here are each bounded and recorded with a replay command, required inputs and their hashes, and each was rechecked by a different actor. They are a sample of what one programme found, not a measure of how often such problems occur. Providers, authors and users can act on them.

Methods in brief

Supplement S1 gives the case selection and its rules. S2 gives the replay and checks for each case, with commands. S3 is the audit trail of withdrawn and narrowed statements.

Roles of AI systems and humans

- **Codex (OpenAI), with GPT-6.1 Sol delegates**
  - took over the programme's database, reference and package work from 28 October;
  - made some original findings, including cases 11 and 12;
  - replayed all twelve cases, built the claims ledger and reviewed this paper's wording.
- **Claude (Anthropic)**, Opus 5.5 in the main session and Claude Sonnet subagents:
  - ran the database, package and reference sessions that made most of the original findings, before 28 October;
  - compiled the initial 189-row candidate catalogue;
  - chose the twelve cases;
  - wrote the interpretation specification and drafted this paper.
- **Barry W. Brook** facilitated the programme, approved the paper on 25 October 2026, chose this case series for development on 8 November 2026, and holds decisions on provider contact, submission and release.

The two AI systems reviewed each other's work throughout. Reviewed snapshots of their correspondence are published in the repository (`docs/correspondence/`), with withheld messages left as placeholders.

Data and code

The repository is public at https://github.com/BWBrook/eprp; code is MIT-licensed and text, figures and data are CC BY-SA 4.0. The claims ledger (`papers/infrastructure-audit/manuscript/claims.csv`) gives, for each proposition, the evidence files, the commit at which they were verified and the replay command. Each replay record names its required inputs and their hashes; some are retained locally (container images, posterior logs, source archives) and are not redistributed. Third-party sources keep their own licences. Records absent from public database snapshots are reported as counts only.

References

Benson, Roger B.J., Roger A. Close, Gawain T. Antell, Robert J. Whittaker, Paul Valdes, Alex Farnsworth, Daniel J. Lunt, Shuzhong Shen, Junxuan Fan, and Erin E. Saupe. 2025. "Marine Animal Diversity Across Latitudinal and Temperature Gradients During the Phanerozoic." *Palaeontology* 68 (3). https://doi.org/10.1111/pala.70006.

Bradshaw, C.J.A., A. Cooper, C.S.M. Turney, and B.W. Brook. 2012. "Robust Estimates of Extinction Time in the Geological Record." *Quaternary Science Reviews* 33: 14–19. https://doi.org/10.1016/j.quascirev.2011.11.021.

Fenton, Isabel S., Adam Woodhouse, Tracy Aze, David Lazarus, Johan Renaudie, Alexander M. Dunhill, Jeremy R. Young, and Erin E. Saupe. 2021. "Triton, a New Species-Level Database of Cenozoic Planktonic Foraminiferal Occurrences." *Scientific Data* 8 (1). https://doi.org/10.1038/s41597-021-00942-7.

Herrero-Gascón, Gloria, Carmen García-Comas, Thomas Bodin, Michael J. Benton, Khushboo Gurung, Benjamin Mills, Alexandre Pohl, and Pedro Cermeño. 2026. "INDITEK -2.0: A Bayesian Inverse Modelling Framework for Mechanistic Evolutionary Processes." *Methods in Ecology and Evolution*. https://doi.org/10.1111/2041-210x.70422.

Matthews, Kara J., Kayla T. Maloney, Sabin Zahirovic, Simon E. Williams, Maria Seton, and R. Dietmar Müller. 2016. "Global Plate Boundary Evolution and Kinematics Since the Late Paleozoic." *Global and Planetary Change* 146: 226–50. https://doi.org/10.1016/j.gloplacha.2016.10.002.

Merdith, Andrew S., Simon E. Williams, Alan S. Collins, Michael G. Tetley, Jacob A. Mulder, Morgan L. Blades, Alexander Young, et al. 2021. "Extending Full-Plate Tectonic Models into Deep Time: Linking the Neoproterozoic and the Phanerozoic." *Earth-Science Reviews* 214: 103477. https://doi.org/10.1016/j.earscirev.2020.103477.

Monarrez, Pedro M., Noel A. Heim, and Jonathan L. Payne. 2023. "Reduced Strength and Increased Variability of Extinction Selectivity During Mass Extinctions." *Royal Society Open Science* 10 (9). https://doi.org/10.1098/rsos.230795.

Rampino, Michael R., and Ken Caldeira. 2025. "Correlation and Cyclicity of Stratigraphic Sequence Boundaries and Chronostratigraphic Stage Boundaries of the Last 253 My: Synchrony of Tectonism, Sea Level, Climate and Biotic Change." *Earth-Science Reviews* 265: 105100. https://doi.org/10.1016/j.earscirev.2025.105100.

Renaudie, Johan, David Lazarus, and Patrick Diver. 2020. "NSB (Neptune Sandbox Berlin): An Expanded and Improved Database of Marine Planktonic Microfossil Data and Deep-Sea Stratigraphy." *Palaeontologia Electronica*. https://doi.org/10.26879/1032.

Silvestro, Daniele, Nicolas Salamin, and Jan Schnitzler. 2014. "PyRate: A New Program to Estimate Speciation and Extinction Rates from Incomplete Fossil Data." *Methods in Ecology and Evolution* 5 (10): 1126–31. https://doi.org/10.1111/2041-210x.12263.

Smith, Jansen A., Elizabeth M. Dowding, Ahmed A. Abdelhady, Paolo Abondio, Ricardo Araújo, Tracy Aze, Mairin A. Balisi, et al. 2025. "Identifying the Big Questions in Paleontology: A Community-Driven Project." *Paleobiology* 51 (3): 408–31. https://doi.org/10.1017/pab.2025.10042.

Strauss, David, and Peter M. Sadler. 1989. "Classical Confidence Intervals and Bayesian Probability Estimates for Ends of Local Taxon Ranges." *Mathematical Geology* 21 (4): 411–27. https://doi.org/10.1007/bf00897326.